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B.S 4
year

PHYS-3701

Electronics-I

Fall-2023

Short Questions:-

- (i) Does P_{dc} increase or decrease with temperature?

P_{dc} & β_{FE} increases with temperature

- (ii) Define avalanche as applied to diode?

Avalanche refers to a process that occurs in a diode when a single carrier triggers a rapid increase in current. This process is also known as avalanche breakdown.

- (iii) Draw characteristic curves of ideal diode model.



- (iv) What is the average value of a full wave rectified voltage with a peak value of 60V?

Average value of a full wave rectified voltage with a peak value of 60V is

$$\frac{2 \times 60V}{\pi} =$$

- (v) Define ripple factor.

The output voltage from rectifier circuit is not steady and it has varying components

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Super imposed on the average value of dc voltage called ripples.

(vi) What is the key characteristics of tunnel diode?

Some key characteristics of Tunnel diode are:

1. Tunnel diode are made from heavily doped p-type and n-type materials.
2. Tunnel diode exhibit negative resistance over part of their forward bias characteristics.
3. Tunnel diode are compact and efficient for specific high-speed and high-frequency applications.
4. The voltage range over which a tunnel diode can be operated properly is 1V or less.

(vii) What are the uses of photodiodes?

Photodiodes are used for:

1. detection of light, both visible and invisible.
2. demodulation and switching.
3. character recognition.
4. optical communication equipment and encoders etc.

(viii) What is the voltage gain of a transistor

1. amplifier that has an output of 5V rms and an input of 250mV rms.

$$A_V = \frac{V_o}{V_i} = \frac{5V_{rms}}{250 \times 10^{-3} V_{rms}}$$

$$A_V = \frac{5V_{rms}}{0.25V_{rms}}$$

$$A_V = 20$$

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- (ix) What are the upper and lower limits of on a dc load line in terms of V_{CE} and I_C ?

The upper limits on dc load line is $I_{C(sat)}$ and $V_{CE(sat)}$. The lower limit is $I_C = 0$ and $V_{CE}(\text{cut off})$.

- (x) What bias conditions must exist for a transistor to operate as an amplifier?

For the normal operation of a Transistor to operate as an amplifier, base emitter junction is forward biased and collector base junction is reverse biased.

- (xi) If a transistor has a $\beta_{dc} = 90$ and emitter resistance = $1k\Omega$, what is the dc input resistance at base?

$$R_B = \beta_{dc} R_E$$

$$R_B = (90)(1k\Omega)$$

$$R_B = 90k\Omega$$

- (xii) How is drain current controlled in JFET?

The drain current in a JFET is controlled by the gate voltage. This is because the gate voltage controls the width of the channel, which in turn controls the flow of current between the source and drain terminals.

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(xiii) What is the main difference in construction of the D-MOSFET and the E-MOSFET?

D-MOSFET

In a MOSFET, the process of removing or depleting the channel of charge carriers and thus decreasing the channel conductivity is called depletion mode.

E-MOSFET

In a MOSFET, the process of creating a channel or increasing the conductivity of the channel by the addition of charge carriers is called enhancement mode.

(xiv) A FET circuit has transconductance of $3000 \mu S$ and $R_d = 10 k\Omega$. Ideally, what voltage gain can it produce?

$$A_v = g_m R_d$$

$$g_m = 3000 \mu S$$

$$R_d = 10 k\Omega$$

$$A_v = +3000 \times 10^{-6} S \times 10 \times 10^3 \Omega$$

$$A_v = 3 \times 10^3 S \times 10 \times 10^3$$

$$A_v = 30$$

(xv) Express a power gain of 50 in decibels.

$$dB = 10 \log_{10}(GP)$$

where $GP = \text{Power gain}$

$$dB = 10 \log_{10}(50)$$

$$\therefore 10 \log_{10} \approx 1.6989$$

$$dB = 10 \times 1.69897 \approx 16.99 dB$$

Long Questions :-

Q#1 (a)

Discuss operation & characteristics of JFET. (5)

From Ilmr book page #127, 128, 129

From MBD page #93 - 96

(b)

A certain JFET has an $I_{GS} = -2\text{nA}$ for $V_{GS} = -20\text{V}$. Find the input resistance. (5)

$$R_{IN} = \left| \frac{V_{GS}}{I_{GS}} \right|$$

$$R_{IN} = \left| \frac{-20\text{V}}{-2\text{nA}} \right|$$

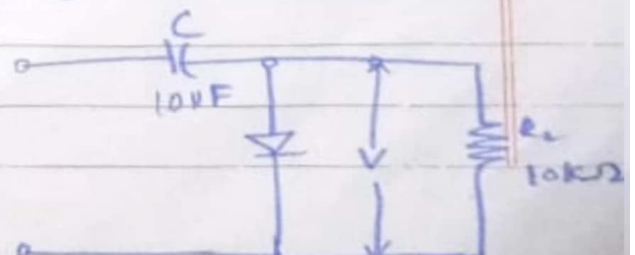
$$R_{IN} = \frac{20\text{V}}{2\text{nA}}$$

$$R_{IN} = 10,000 \text{ M}\Omega$$

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Q#2 (a)

What is the output voltage that you would expect to observe across load resistance in the clamping circuit of figure 1. Assume that RC is large enough to prevent significant capacitor discharge. Also draw the expected waveform. (5)



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Ideally, a negative dc value equal to the input less the diode drop is inserted by the clamping circuit.

$$V_{DC} \approx -(V_{p(\text{in})} - 0.7V)$$

$$V_{DC} \approx -(24V - 0.7V)$$

$$V_{DC} \approx -23.3V$$

Actually, the capacitor will discharge slightly between peaks, and, as a result, the output voltage will have an average value of slightly less than calculated above. The output waveform goes to approximately $+0.7V$, as shown in figure below.



output waveform across R_L

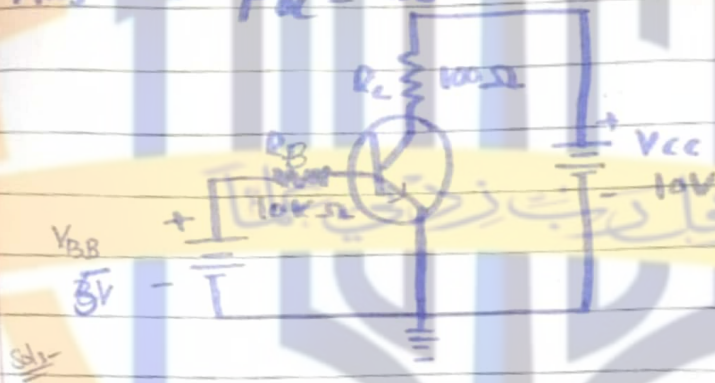
(b)

Briefly explain construction & working principle of Schottky diode. Also discuss its applications. (5)

From Ilmi book page # 272

Q#3 (a)

Determine I_B , I_C , I_E , V_{CE} and V_{CB} in the circuit of figure 2. The transistor has a $\beta_{DC} = 150$. (6)



when the base-emitter junction is forward biased, it is likely a forward biased diode and has a nominal forward voltage drop of $V_{BE} \approx 0.7V$

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{5V - 0.7V}{10K\Omega} = 430\mu A$$

$$I_C = \beta_{DC} I_B = (150)(430\mu A) = 64.5mA$$

Solve for V_{CE} and V_{CB} .

$$V_{CE} = V_{CC} - I_C R_C$$

$$V_{CE} = 10V - (64.5mA)(100\Omega)$$

$$V_{CE} = 10V - 6.45V$$

$$V_{CE} = 3.55V$$

$$I_E = I_C + I_B = 64.5 \text{ mA} + 430 \mu\text{A}$$

$$I_E = \underline{64.9 \text{ mA}}$$

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$$V_{CB} = V_{CE} - V_{BE}$$

$$V_{CB} = 3.55 \text{ V} - 0.7 \text{ V}$$

$$V_{CB} = \underline{2.85 \text{ V}}$$

$$I_B = 430 \mu\text{A}, I_C = 64.5 \text{ mA}, I_E = 64.9 \text{ mA}, V_{CE} = 3.55 \text{ V}, V_{CB} = 2.85 \text{ V}$$

Since collector is at higher voltage than the base, the collector-base junction is reverse-biased.

(b).

Discuss how a transistor is used as an electronic switch. (4)

From Electronic devices by Floyd 9th edition
page #192 Topic (4.5)

وقل رب زدني علما

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